

**Course Code: 305-05**  
**Course Title: Web Designing-I**

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| Course Code                                                             | 305-05                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |      |      |      |      |      |      |      |      |  |
| Course Title                                                            | Web Designing-I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |      |      |      |      |      |      |      |  |
| Credits                                                                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |      |      |      |      |      |      |      |  |
| Course Category                                                         | Major Course                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |      |      |      |      |      |      |      |  |
| Level of Course                                                         | 300-399 ( Higher Level )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |      |      |      |      |      |      |      |      |  |
| Teaching per Week                                                       | 2 Hours Theory + 4 Hours Practical work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |      |      |      |      |      |      |      |  |
| Minimum weeks per Semester                                              | 15 (Including class work, examination, preparation etc.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |      |      |      |      |      |      |      |      |  |
| Review / Revision                                                       | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |      |      |      |      |      |      |      |  |
| Implementation Year:                                                    | A.Y. 2025-2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |      |      |      |      |      |      |      |      |  |
| Purpose of Course                                                       | <ul style="list-style-type: none"><li>- Design is the process of collecting ideas, and aesthetically arranging and implementing them, guided by certain principles for a specific purpose.</li><li>- Web design is a similar process of creation, with the intention of presenting the content on electronic web pages, which the end- users can access through the internet with the help of a web browser.</li><li>- This course deals with designing of websites.</li></ul>                                                                                                                                                                                                                                                               |      |      |      |      |      |      |      |      |  |
| Course Objective                                                        | To make students aware of web terminology and website designing tools. Student can understand and implement the real functions of website development.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |      |      |      |      |      |      |      |      |  |
| Pre-requisite                                                           | Basic knowledge of Simple HTML and HTML-5 concepts, windows based applications. Some very basic acquaintance with computers and the www is assumed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |      |      |      |      |      |      |      |  |
| Course Outcomes                                                         | <b>CO1:</b> Develop proficiency in HTML5 syntax and semantics, and CSS styling techniques for creating visually appealing web pages.<br><b>CO2:</b> Implement Bootstrap framework for rapid prototyping and responsive design, ensuring cross-browser compatibility and scalability.<br><b>CO3:</b> Utilize JavaScript for interactivity and dynamic content manipulation, incorporating libraries like jQuery for streamlined development.<br><b>CO4:</b> Demonstrate the ability to integrate HTML5, CSS, Bootstrap, and JavaScript to create cohesive and engaging web applications.<br><b>CO5:</b> Apply industry-standard practices in web development, including code optimization, version control, and responsive design principles. |      |      |      |      |      |      |      |      |  |
| Mapping between Course Outcomes(CO) with Program Specific Outcomes(PSO) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | PSO1 | PSO2 | PSO3 | PSO4 | PSO5 | PSO6 | PSO7 | PSO8 |  |
|                                                                         | CO1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |      |      |      |      |      |      |      |  |
|                                                                         | CO2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |      |      |      |      |      |      |      |  |
|                                                                         | CO3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |      |      |      |      |      |      |      |  |
|                                                                         | CO4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |      |      |      |      |      |      |      |  |
|                                                                         | CO5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |      |      |      |      |      |      |      |  |
| Course Content                                                          | <b>UNIT 1: HTML5 and CSS for Web Design</b><br>1.1 Concepts of CSS:<br>1.1.1 Adding CSS (Inline, Internal, External)<br>1.1.2 HTML Links and attributes ( _self, _blank, _parent, _top)<br>1.1.3 Absolute URL and Relative URL in <href><br>1.1.4 <img> tag and its attributes (src, alt, style, width, height)<br>1.2 HTML Forms:<br>1.2.1 Form Elements and Their Attributes:<br>1.2.1.1 <form> (action, method, novalidate, autocomplete, target)<br>1.2.1.2 <label>, <input> (text, radio, checkbox, submit/reset button)<br>1.2.1.3 <select> (id, name), <option><br>1.2.1.4 <textarea> (name, rows, cols)<br>1.2.1.5 <button> (type, onclick)                                                                                          |      |      |      |      |      |      |      |      |  |

- 1.2.1.6 <datalist>
- 1.2.2 Media: Video and Audio

## **UNIT 2: Responsive Design Using Bootstrap 4**

- 2.1 Bootstrap Introduction
- 2.2 Grid Structure
- 2.3 Table, Colours, Alerts, Form Controls
- 2.4 Buttons and Button Groups
- 2.5 Images, Media Objects
- 2.6 Pagination
- 2.7 Bootstrap Grids
- 2.8 Bootstrap Themes

## **UNIT 3: JavaScript Basics and Control Structures**

- 3.1 Overview of Client & Server-Side Scripting
- 3.2 Structure of JavaScript
- 3.3 Data Types and Variables
- 3.4 Operators:
  - 3.4.1 Arithmetic, Assignment, Comparison
  - 3.4.2 Logical and Conditional Operators
- 3.5 Control Structures:
  - 3.5.1 if...else, switch...case
  - 3.5.2 while, do...while, for loop
  - 3.5.3 break, continue
- 3.6 JavaScript Strings and Events:
  - 3.6.1 JavaScript String Types
  - 3.6.2 String Functions:
    - concat(), split(), indexOf(), lastIndexOf(), substring(), trim(), slice(), replace(), charAt()
  - 3.6.3 JavaScript Events:
    - 3.6.3.1 Mouse Events: click, mouseover, mouseout, mouseup
    - 3.6.3.2 Keyboard Events: keyup, keydown
    - 3.6.3.3 Form Events: focus, submit, blur, change

## **UNIT 4: Advanced JavaScript and DOM Manipulation**

- 4.1 JavaScript Objects:
    - 4.1.1 Creating Object (By literal, instance, constructor)
  - 4.2 Date Object:
    - 4.2.1 Date Constructor: Date(), Date(milliseconds), Date(dateString), Date(year, month, day, hours, minutes, seconds, milliseconds)
    - 4.2.2 Date Methods: getDate(), getDay(), getMonth(), getHours(), setDate(), setMonth(), setDay(), toString()
  - 4.3 JavaScript Functions:
    - 4.3.1 Defining Functions (with and without parameters)
    - 4.3.2 Calling Functions
    - 4.3.3 Return Statement
    - 4.3.4 Page Redirection
  - 4.4 Dialog Boxes and Form Validation:
    - 4.4.1 Dialog Boxes: alert, confirm, prompt
    - 4.4.2 Form Validation:
      - 4.4.2.1 Basic Validation (All form details are filled)
      - 4.4.2.2 Data Format Validation (email, number, string, mobile number, name)
- [All Units carry Equal Weightage]

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| <b>Reference Books</b>      | <ol style="list-style-type: none"> <li>1. HTML &amp; CSS: The Complete Reference - Thomas Powell - McGraw Hill Education</li> <li>2. HTML Unleashed, Darnell Rick –Techmedia</li> <li>3. HTML, XHTML, and CSS Bible - Steven M. Schafe – Wiley Publications</li> <li>4. Cascading Style Sheets- The Definitive Guide, E. A Meyer –O'Reilly</li> <li>5. Java Scripting Programming for Absolute Beginner, Harris -PHI</li> <li>6. JavaScript Step by Step, Suehring -PHI</li> <li>7. Bootstrap in 24 Hours, Sams Teach Yourself - JenniferKyrnin</li> <li>8. Learning Bootstrap 4 - Matt Lambert – Packt Publishing</li> <li>9. Bootstrap Responsive Web Development - Jake Spurlock - O'Reilly Media.</li> <li>10. JavaScript and JQuery (Interactive Front-End Web Development) by Jon Duckett</li> <li>11. JavaScript and JQuery (The missing manual) by David Sawyer MCFarland</li> </ol> |
| <b>Teaching Methodology</b> | Class Work, Discussion, Lab work, Self-Study, Seminars and/or Assignments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Evaluation Method</b>    | <p>50% Internal assessment.</p> <p>50% External assessment.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |